3-manifold invariants from the Hopf algebras $\mathbb{F}_p[x]/(x^p)$ and $\mathbb{R}[x]$

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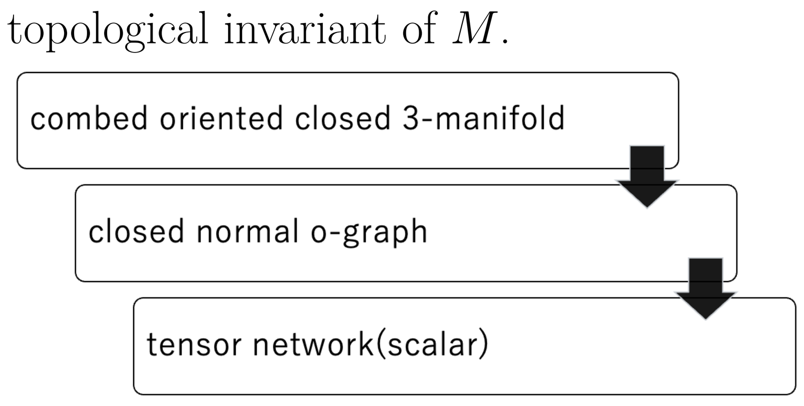
Throughout, an o-graph means a closed normal o-graph and 3-manifold means a closed oriented 3-manifold.

Introduction

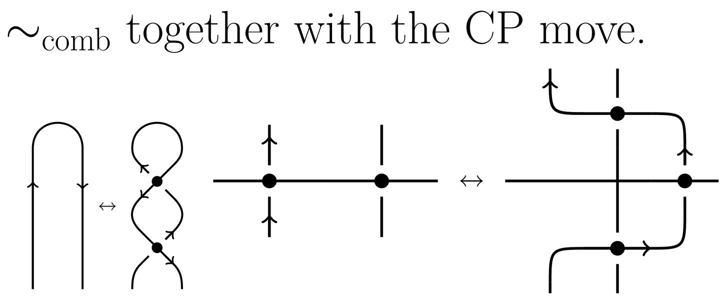
Mihalache–Suzuki–Terashima (MST) construct an invariant $Z(M, v; H)$ of a closed oriented 3-manifold M with a combing v (nonvanishing vector field) from a finite-dimensional involutive (i.e. the antipode is an involution) Hopf algebra H over a field $\Bbbk$ via an o-graph Γ representing (M, v) [MST24]. If H is semisimple, then the invariant is independent of the combing v, hence defines a topological invariant of M.



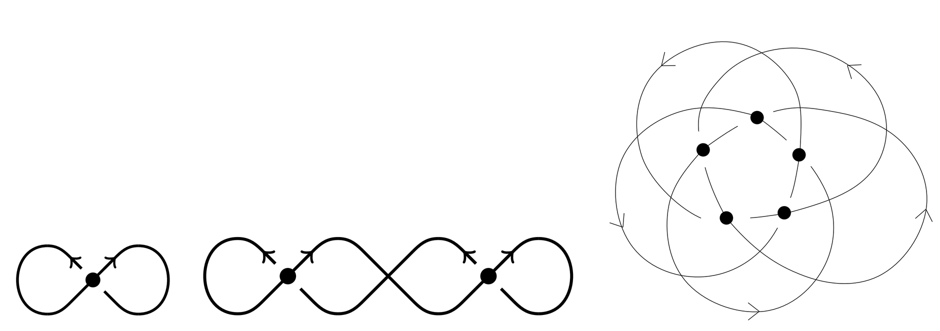
In our specialization, we take $\Bbbk = \mathbb{F}_p$ and $H = \mathbb{F}_p[x]/(x^p)$, and derive an explicit formula in terms of the first homology group $H_1(M)$. We also modify the construction for $H = \mathbb{R}[x]$ and determine the corresponding invariant.

Topological Foundations

A closed normal o-graph (Benedetti–Petronio) is a virtual oriented link diagram encoding a branched flow-spine of a closed oriented 3-manifold M with a combing v[BP97]. The equivalence relation $\sim_{\text{comb}}$ on o-graphs representing the same (M, v) is generated by planar isotopy, 0–2 moves, and MP moves. Moreover, the equivalence relation $\sim_{\text{homeo}}$ on o-graphs representing the same M is generated by $\sim_{\text{comb}}$ together with the CP move.



0–2 move and a portion of MP move



Examples of o-graphs

Algebraic foundations

A Hopf algebra is a tuple $H = (H, M, 1_H, \Delta, \varepsilon, S)$ consisting of a vector space H, an algebra $(H, M, 1_H)$, a coalgebra (H, Δ, ε), and an antipode S satisfying certain compatibility axioms.

Let $A := \mathbb{F}_p[X] := \mathbb{F}_p[x]/(x^p)$. Let $X = x + (x^p) \in A$. Then X is primitive, i.e.

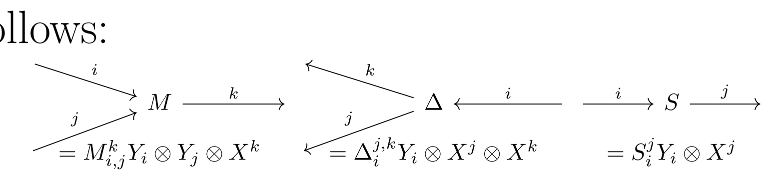
$$\Delta(X) = X \otimes 1 + 1 \otimes X, \ \ \varepsilon(X) = 0, \ \ S(X) = -X.$$

$(X^n)_{n \in J}$, $J = \{0, 1, \ldots, p-1\}$ is a basis of A and let $(Y_n)_n \in A^*$ be its dual basis.

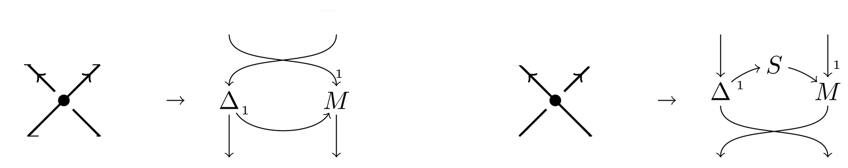
Let $B = \mathbb{R}[x]$, where x is primitive. Both A and B are involutive Hopf algebras. The algebra A is finite-dimensional and (co)unimodular, whereas B is infinite-dimensional and has no nonzero (co)integral.

MST construction

Let us explain the construction of $Z(M; A)$ for $A = \mathbb{F}_p[X]$ concretely. A **tensor network** on a vector space A is an oriented graph (V, E) representing a tensor T on A. Each vertex represents a tensor, and each incoming (resp. outgoing) edge specifies covariant (resp. contravariant) component of the tensor of the vertex. $s, t : E \to V$ denote source and target maps of the graph. Contracting tensors is represented by connecting edges. Let E be the set of edges. We introduce projection $T(c)$ of T according to states $c \in J^E$ of Γ, such that each component specified by an edge is projected to a basis element specified by the state. For example, the tensors M, Δ, S are represented as follows:



The invariant $Z(M; A)$ is the resulting tensor replacing each vertex of Γ as shown below:



Replace each vertex by the corresponding tensor

Construction for $\mathbb{R}[x]$

To apply the MST construction to infinite-dimensional B, we introduce Weyl-algebraic calculus. Let $\mathbb{W} := \mathbb{R}\langle x, \partial_x \rangle / (\partial_x x - x\partial_x = 1)$ be the Weyl algebra, $\mathbb{W}_g := \mathbb{W}[[g]]$ the Rees Weyl algebra. We set elements in $\mathbb{W}_g^{\otimes 2}$

$$T = \exp(g\partial_x \otimes x), \quad \bar{T} = \exp(-g\partial_x \otimes x)$$

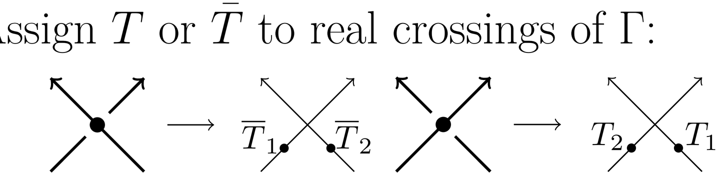
We introduce a formal trace Tr on the Weyl algebra by

$$\mathrm{Tr}\left(\sum_{n=0}^{\infty} \frac{a^n}{n!} x^n \partial_x^n\right) := -\frac{1}{a} \qquad (a \in \mathbb{R}((g)) \setminus \{0\})$$

Note that its domain is enough for our purpose, and the trace is cyclic.

The invariant of (M, v) is constructed via an o-graph Γ representing (M, v) as follows:

1. Assign T or $\bar{T}$ to real crossings of Γ:



2. Cut Γ at one point, called the **basepoint**, on an edge of Γ and take the product of all elements from the basepoint along the strand of Γ.
3. Take the trace of the resulting element u and take the limit.

$$Z(M, v; \mathbb{R}[x]) = \lim_{g \to 1} \mathrm{Tr}(u).$$

Main Result (for $H = \mathbb{F}_p[X], \mathbb{R}[x]$)

For any closed oriented 3-manifold M with a combing v, we have invariants

$$Z(M; \mathbb{F}_p[X]) = \begin{cases} 0 & (|H_1(M)| = \infty \text{ or } p \mid |H_1(M)|), \\ 1 & (\text{otherwise}). \end{cases}$$

$$Z(M, v; \mathbb{R}[x]) = \pm 1/|H_1(M)| \qquad (|H_1(M)| < \infty, \ \pm \text{ depends on } v).$$

Proof sketch for $H = \mathbb{F}_p[X]$

This was conjectured by T.Kawakami [Kaw23]. Since $\mathbb{F}_p[X]$ is unimodular and counimodular, $Z(M; \mathbb{F}_p[X])$ is a topological invariant of M.

Step 1:construct isomorphism

Set $\exp^{[p]}(z) := \Sigma_{n=0}^{p-1} \frac{z^n}{n!}$. Then,

$$\phi : \mathbb{F}_p[\mathbb{Z}/p\mathbb{Z}] \to \mathbb{F}_p[X], g \mapsto \exp^{[p]}(X).$$

is an isomorphism of algebras but not of Hopf algebras, where g is a generator of $\mathbb{Z}/p\mathbb{Z}$.

Step 2: introduce flow of graph

For a colored tensor network (Γ, c), we define the net flow at each vertex by

$$N_{(\Gamma,c)}(v) = \Sigma_{i \in s^{-1}(v)} c_i - \Sigma_{o \in t^{-1}(v)} c_o \in \mathbb{Z}.$$

Step 3: calculate flow of D

To compare the coalgebra structures, we set

$$D = (\phi \otimes \phi)\Delta_{\mathbb{F}_p[\mathbb{Z}/p\mathbb{Z}]}\phi^{-1} - \Delta_{\mathbb{F}_p[X]} : H \to H^{\otimes 2}.$$

We consider the colored diagram of D. A concrete calculation shows that the flow of the vertex of D is negative if $D(c) \neq 0$.

Step 4: kill the term D Since the tensor network of $Z(M; \mathbb{F}_p[X])$ is closed as a graph, the sum of net flows over all vertices vanishes. If we only consider states c such that $D(c) \neq 0$, then the flow of vertices M, S is 0 but D is negative. Thus, the term D does not contribute to the invariant:

$$Z(M; \mathbb{F}_p[X]) = Z(M; \mathbb{F}_p[\mathbb{Z}/p\mathbb{Z}]).$$

Step 5: apply formula for $\Bbbk[G]$

In [MST24], they show that for a finite group G,

$$Z(M; \mathbb{C}[G]) = |\operatorname{Hom}(\pi_1(M), G)|.$$

Finally, we see that

$$Z(M; \mathbb{F}_p[X]) \equiv |\operatorname{Hom}(H_1(M), \mathbb{Z}/p\mathbb{Z})| \pmod p$$

Proof sketch for $H = \mathbb{R}[x]$

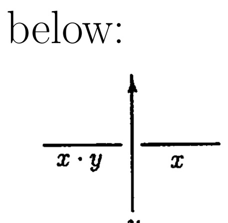
The parameter g is introduced as a regularization parameter to justify infinite sum.

Value of the invariant

Let N be the number of real crossings of Γ, and let $B \in \mathbb{Z}^{N \times N}$ be the matrix such that $B_{i,j} = \sigma$ if the i-th strand goes over the j-th strand at a crossing with sign σ, and $B_{i,j} = 0$ otherwise. We set $\omega_N = \Sigma_{i=1}^{N}(e_{i,i} - e_{i+1,i}) \in \mathbb{Z}^{N \times N}$, where $e_{i,j}$ is the matrix unit and $e_{N+1,N} = e_{1,N}$. A formula for the trace yields

$$Z(M, v; \mathbb{R}[x]) = \lim_{g \to 1} \frac{-1}{\det(\omega_N + B)}.$$

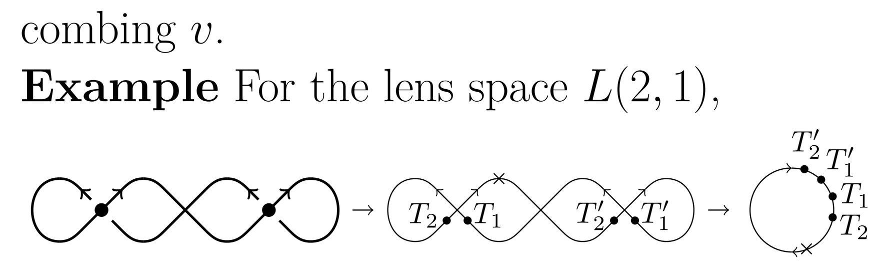
On the other hand, $pi_1(M)$ is isomorphic to the group generated by arcs of Γ with relations given as shown below:



We see that $A = \omega_N + B$ is the presentation matrix of $H_1(M)$, so $\det A = \pm|H_1(M)|$.

Remarks on invariance Since the trace is cyclic, the choice of basepoint does not change the value. Thus $Z(M, v; \mathbb{R}[x])$ is an invariant of (M, v). However, the CP move changes the sign of u, so $Z(M, v; \mathbb{R}[x])$ depends on the combing v.

Example For the lens space $L(2, 1)$,



$$\begin{aligned}
&\lim_{g \to 1} \mathrm{Tr}\, T_2' T_1' T_1 T_2 \\
&= \lim_{g \to 1} \mathrm{Tr} \sum_{n,m=0}^{\infty} \frac{1}{n!\, m!} x^m \partial_x^m \partial_x^n x^n \\
&= \lim_{g \to 1} \mathrm{Tr}\, 2^{x\partial_x}(1-g)^{-x\partial_x - 1} = -1/2 \\
&= -1/\det([1, 1, -1; -1, 1, 0; 0, 0, 1])
\end{aligned}$$

Applications of methods

Van der Veen and Bar-Natan found that the Alexander polynomial is obtained as the inverse of the universal invariant of a knot K with a certain representation ρ of the Drinfeld double of the Borel half $U = U(\mathfrak{sl}_2^+)$[BV21]:

$$Z(K; \mathcal{D}(U); \rho) = 1/\Delta_K(t)$$

They essentially construct the representation ρ by Weyl algebraic calculus. We expect that other invariants using infinite-dimensional algebras can be obtained by similar methods.

Future Directions

Non-involutory generalizations

In order to define invariants from non-involutory Hopf algebras, we need to take the framing of the manifold and the knot into account. In Benedetti-Petronio's work, the framing of the manifold is encoded by an o-graph with $\mathbb{Z}_2$-weights on edges. Besides, a balancing element is needed to attach to the knot.

Extension of intertwiner property

The universal invariant of the usual tangle diagram is an intertwiner in its representation space. This is essential for defining the value of certain invariants, such as the colored Jones polynomial via Schur's lemma.

However, invariants of tangles of o-graphs do NOT have such a property.

Twisted version

The twisted Drinfeld double has been defined and related twisted versions of knot invariants have been studied. However, the twisted version of the Heisenberg double, in Turaev's sense, has not been defined yet. Partially, a twisted invariant of closed 3-manifolds has been defined using o-graphs with group-weights.

Also, the absolute Reidemeister torsion of closed 3-manifolds has not been obtained in this framework.

Relation with Kuperberg invariants

References

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